Tangent Lévy v2
Risk-Neutral Drift Audit Summary

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Abstract

This note records the final v2 audit of a reduced Tangent Lévy construction under the risk-neutral measure Q. The experiment uses the native Tangent Lévy short-end spot compensator, a deterministic Musiela shift, and a restricted universe consisting of spot plus 44 fixed-cash-strike options. An exact Musiela/compound-Poisson decomposition verifies the unconditional martingale restriction for all 570 active day–instrument coordinates, and the two conditional drift tests do not reject it. The literal coordinate-wise Gaussian diagnostic reports extreme statistics because some compensating jumps are too rare to be sampled by 100,000 paths; it is retained as a Monte Carlo resolution diagnostic rather than used as the no-arbitrage decision. The result concerns this reduced Q construction, not a successful calibration of the full stochastic model in the paper.

1 Scope and implementation

The population contains 100,000 simulated paths (50,000 antithetic pairs), initialized from the 1 June 2022 SPX surface. The tested trading universe is spot plus 44 options: eleven normalized-moneyness instruments at each of 5, 10, 15, and 20 business-day maturities. Strikes remain fixed in cash units over each one-step return. Spot is advanced with the Tangent Lévy codebook's short-end compensator, while the codebook is transported by deterministic Musiela shift substeps.

The implementation includes the finite-activity no-jump atom in transform pricing, lattice-consistent FFT pricing, removal of numerical FFT dust, tail cutoffs under both cash and share intensities, short-end midpoint evaluation in Musiela substeps, float64 audit returns, and exact fold-level entropy scoring. These details are necessary for the simulated spot and option prices to be internally consistent under Q.

2 Calibration limitation

The attempted historical reconstruction did not pass its mandatory gate on any of 335 dates. Across the history the maximum relative L^1 error was 1.3817 and the maximum repricing RMSE was 525 quoted half-spreads. For the retained 1 June 2022 initialization, the relative L^1 error is 0.9473, the repricing RMSE is 20.185 half-spreads, and the maximum absolute normalized price error is 0.00977. This confirms the expected poor fit.

Because the reconstructed history supplied no positive covariance mode, the final audit uses zero stochastic factors. It is a deterministic-codebook Tangent Lévy model, not a fitted version of the paper's full stochastic factor/HJM specification. The drift-test methodology is adapted from `paper/main2.tex`; that paper's underlying SDLV experiment uses a Gaussian/information-bridge spot construction and does not by itself calibrate the sampling distribution of a rare-event compound-Poisson diagnostic.

3 Final Q results

For the deterministic codebook, the unconditional restriction can be checked without simulation. At every day and remaining maturity, the integrated Lévy measure in today's option price must equal the one-day short-end measure plus tomorrow's shifted integrated measure. Equality holds componentwise on the 301-node jump grid. It therefore gives equality of the full characteristic exponents and establishes the one-step martingale identity for every European strike, not merely the 44 strikes retained in the audit. Across the 50 active day–maturity decompositions, the maximum relative density error is 3.004×10^{-16}; the maximum spot first-moment error is 1.110×10^{-16}. Both are below the 10^{-12} audit tolerance, so the exact unconditional audit is a **PASS** over all 570 active coordinates.

Test	Mean score	Pair SE	t statistic	95% interval	Result
Conditional, squared	3.340×10^{-8}	7.299×10^{-8}	0.458	$[-1.096, 1.764] \times 10^{-7}$	**PASS**
Conditional, entropy	1.702×10^{-9}	2.915×10^{-9}	0.584	$[-4.012, 7.415] \times 10^{-9}$	**PASS**

Table 1: Cross-fitted conditional drift tests on the retained population.

Both conditional confidence intervals contain zero and both absolute t statistics are below 1.96. Accordingly, neither conditional test rejects the null of zero predictable drift at the 5% two-sided level.

The literal unconditional report tests 570 time–instrument coordinates. It finds a maximum absolute z statistic of 1.485×10^9, a 99th percentile of 1.196×10^8, 204 coordinates above 1.96, 150 above 2.576, and 138 above 5. Read literally as Gaussian tests, these coordinates fail.

That literal reading is not statistically valid for the problematic coordinates. Some reconstructed annual intensities are approximately 1.5×10^{-6}, corresponding to a one-day jump probability near 5.95×10^{-9}. A 100,000-path population is therefore overwhelmingly likely to contain no compensating jump. The empirical variance collapses while the analytically compensated mean remains nonzero, producing enormous studentized ratios. On the 432 coordinates with pair standard error exceeding 10^{-12}, the maximum absolute statistic is 3.597, the 99th percentile is 3.551, 66 exceed 1.96, 12 exceed 2.576, and none exceeds 5. This filtered view is diagnostic only; it is not a replacement preregistered test.

4 Conclusion

The exact unconditional audit confirms the Q martingale restriction for the reduced deterministic-codebook construction, and the two requested conditional tests find no detectable predictable drift. The Gaussian coordinate screen cannot reach the same conclusion at this Monte Carlo size because the fitted finite-activity model contains events far rarer than the population can resolve. Its extreme statistics are a rare-event resolution failure, not evidence against the exact martingale identity.

This audit does not validate the market fit, the physical-measure dynamics, or the omitted stochastic codebook factors. It establishes that the implemented reduced construction uses the Tangent Lévy compensator consistently and that its two cross-fitted conditional drift tests pass on the retained population.

5 Retained artifacts

The canonical result root is `/home/hans/quantdata_data/training/dysanos/tangent-levy/v2/spx-891-v2`. It retains the 1 June 2022 initialization checkpoint and these current result families:

- `q-reduced-population/` arrays;
- `q-reduced-audit.json` Monte Carlo diagnostics;
- `q-reduced-analytic.json` exact unconditional audit; and
- both conditional JSON reports and their cross-fit checkpoints.

Superseded daily reconstruction checkpoints and legacy aggregate calibration logs are intentionally excluded.